Filter

on Lie Groups

for GNSS/INS Tightly-Coupled Navigation

Technical Documentation

State Group:

$\mathcal{G} = \mathrm{SE}_2(3) \times T(6)$

State dimension: $p = 15$ **Sigma points:** $2L + 1 = 67$

Implementation Reference Manual

Scientific Initiation Project

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1 Introduction

Inertial Navigation Systems (INS) fused with Global Navigation Satellite Systems (GNSS) constitute the backbone of modern navigation. Classical Kalman filtering approaches parametrise the vehicle attitude with Euler angles or quaternions and propagate the covariance in $\mathbb{R}^n$, ignoring the underlying non-Euclidean geometry of the state space. This leads to linearisation errors and singularities near gimbal-lock.

The *UKF on Lie Groups* (UKF-Lie) addresses this by working directly on a matrix Lie group $\mathcal{G}$, using the group exponential and logarithm to map between the manifold and its tangent space (the Lie algebra) only when computing means and covariances. The sigma-point propagation itself remains entirely on the group, preserving the rotation-matrix structure at every step.

1.1 Notation

Table 1: Symbol notation.

Symbol	Dimension	Description
$\mathbf{C}_{eb} \in \mathrm{SO}(3)$	3×3	Rotation matrix: body $\to$ ECEF
$\mathbf{v}^e_{eb} \in \mathbb{R}^3$	3×1	Velocity of body in ECEF frame
$\mathbf{p}^e_{eb} \in \mathbb{R}^3$	3×1	Position of body in ECEF frame
$\mathbf{b}_a \in \mathbb{R}^3$	3×1	Accelerometer bias
$\mathbf{b}_g \in \mathbb{R}^3$	3×1	Gyroscope bias
$\mathbf{f}^b_{ib} \in \mathbb{R}^3$	3×1	Specific force (accelerometer measurement)
$\boldsymbol{\omega}^b_{ib} \in \mathbb{R}^3$	3×1	Angular rate (gyroscope measurement)
$\mathbf{X} \in \mathcal{G}$	13×13	Full navigation state on the Lie group
$\mathbf{P} \in \mathbb{R}^{p \times p}$	15×15	Error covariance in the Lie algebra
$\mathbf{Q}$	15×15	Process noise covariance
$\mathbf{R}$	3×3	Measurement noise covariance
$L = 2p + q$	33	Augmented state dimension
α, β, κ	scalars	UT scaling parameters
λ	scalar	$\alpha^2(L+\kappa) - L$

2 Lie Group Fundamentals

2.1 The Special Orthogonal Group SO(3)

Definition 2.1 (SO(3))**.** The special orthogonal group:

$$\mathrm{SO}(3) = \{\mathbf{C} \in \mathbb{R}^{3\times 3} \mid \mathbf{C}^\top \mathbf{C} = \mathbf{I},\ \det(\mathbf{C}) = +1\}.$$

For $\boldsymbol{\phi} = \phi\,\mathbf{a}$, the hat operator: $[\boldsymbol{\phi}]_{\times ij} = -\epsilon_{ijk}\phi_k$.

Exponential map (Rodrigues formula):

$$\mathrm{Exp}(\boldsymbol{\phi}) = \mathbf{I} + \frac{\sin\phi}{\phi}[\boldsymbol{\phi}]_\times + \frac{1-\cos\phi}{\phi^2}[\boldsymbol{\phi}]^2_\times.$$

Logarithmic map:

$$\phi = \arccos\!\left(\tfrac{\mathrm{tr}(\mathbf{C})-1}{2}\right), \qquad \mathbf{a} = \frac{1}{2\sin\phi}\left(\mathbf{C} - \mathbf{C}^\top\right)^\vee, \qquad \mathrm{Log}(\mathbf{C}) = \phi\,\mathbf{a}.$$

2.2 The Extended Pose Group SE2(3)

Definition 2.2 ($\mathrm{SE}_2(3)$)**.**

$$\mathrm{SE}_2(3) = \left\{ \begin{pmatrix} \mathbf{C} & \mathbf{v} & \mathbf{p} \\ \mathbf{0}_{2\times 3} & \mathbf{I}_2 & \end{pmatrix} \in \mathbb{R}^{5\times 5} \;\middle|\; \mathbf{C} \in \mathrm{SO}(3),\ \mathbf{v}, \mathbf{p} \in \mathbb{R}^3 \right\}.$$

Exponential map (`exp_multiSE3.m`):

$$\mathrm{Exp}(\xi) = \mathbf{I}_5 + \Xi + \frac{1-\cos\theta}{\theta^2}\,\Xi^2 + \frac{\theta - \sin\theta}{\theta^3}\,\Xi^3, \quad \theta = \|\boldsymbol{\phi}\|.$$

Logarithmic map (`log_multiSE3.m`):

$$\mathrm{Log}(\mathbf{X}) = \begin{pmatrix} \phi\,\mathbf{a} \\ \mathbf{J}^{-1}(\phi)\,\mathbf{r}_v \\ \mathbf{J}^{-1}(\phi)\,\mathbf{r}_p \end{pmatrix}, \qquad \mathbf{J}^{-1}(\phi) = \frac{\phi}{2}\cot\frac{\phi}{2}\,\mathbf{I} + \left(1 - \frac{\phi}{2}\cot\frac{\phi}{2}\right)\mathbf{a}\mathbf{a}^\top - \frac{\phi}{2}[\mathbf{a}]_\times.$$

2.3 Full State Group

The 13×13 block-diagonal navigation state:

$$\mathbf{X} = \mathrm{blkdiag}\!\left(\begin{pmatrix} \mathbf{C}_{eb} & \mathbf{v}^e_{eb} & \mathbf{p}^e_{eb} \\ \mathbf{0} & \mathbf{I} & \end{pmatrix}, \begin{pmatrix} \mathbf{I} & \mathbf{b}_a \\ \mathbf{0} & 1 \end{pmatrix}, \begin{pmatrix} \mathbf{I} & \mathbf{b}_g \\ \mathbf{0} & 1 \end{pmatrix} \right) \in \mathbb{R}^{13\times 13}.$$

Adjoint map (`Ad_G.m`):

$$\mathrm{Ad}_{\mathbf{X}} = \begin{pmatrix} \mathbf{C}_{eb} & \mathbf{0} & \mathbf{0} \\ [\mathbf{v}^e_{eb}]_\times \mathbf{C}_{eb} & \mathbf{C}_{eb} & \mathbf{0} \\ [\mathbf{p}^e_{eb}]_\times \mathbf{C}_{eb} & \mathbf{0} & \mathbf{C}_{eb} \end{pmatrix} \oplus \mathbf{I}_6.$$

3 Navigation Model

3.1 Reference Frames

Rotation NED to ECEF (`DCM_en.m`):

$$\mathbf{C}_{en}(\lambda, \ell) = \begin{pmatrix} -\sin\lambda\cos\ell & -\sin\ell & -\cos\lambda\cos\ell \\ -\sin\lambda\sin\ell & \cos\ell & -\cos\lambda\sin\ell \\ \cos\lambda & 0 & -\sin\lambda \end{pmatrix}.$$

WGS-84 gravity (`gravityModel.m`):

$$g(\lambda) = \frac{g_0\,(1 + g_1\sin^2\lambda)}{\sqrt{1 - e^2\sin^2\lambda}}, \quad g_0 = 9.7803267714\,\mathrm{m/s^2}, \quad g_1 = 1.93185 \times 10^{-3}, \quad e = 0.0818.$$

3.2 Process Model

Left-invariant continuous-time dynamics (`Omega.m`):

$$\dot{\mathbf{X}} = \mathbf{X}\,\Omega^\wedge, \qquad \Omega = \begin{pmatrix} \boldsymbol{\omega}^b_{ib} - \mathbf{b}_g \\ \mathbf{C}_{be}(g_n \mathbf{f}^b_{ib} - \mathbf{b}_a) + \mathbf{C}_{be}\mathbf{g}^e \\ \mathbf{C}_{be}\mathbf{v}^e_{eb} \\ \mathbf{0}_6 \end{pmatrix} \in \mathbb{R}^{15}.$$

Discrete integration:

$$\mathbf{X}_k = \mathbf{X}_{k-1}\,\mathrm{Exp}(\Omega_{k-1}\,\Delta t + \boldsymbol{q}_{k-1}), \quad \boldsymbol{q}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}).$$

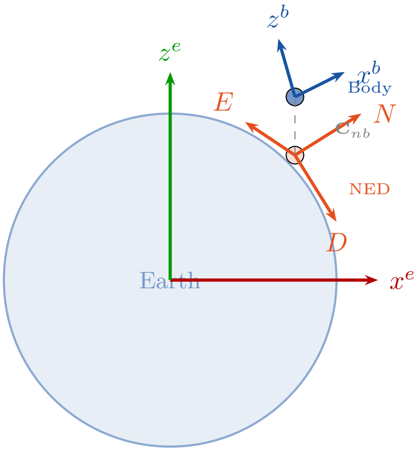

Figure 1: Reference frames: ECEF (red/green), local NED (orange), and body frame (blue).

3.3 Measurement Model

GNSS position with lever arm compensation:

$$h(\mathbf{X}) = \mathbf{p}_{eb}^{e} + \mathbf{C}_{eb}\,\boldsymbol{\ell}_b \in \mathbb{R}^3, \qquad Y = \begin{pmatrix} \mathbf{I}_3 & h(\mathbf{X}) \\ \mathbf{0} & 1 \end{pmatrix} \in T(3).$$

Innovation (since $T(3)$ is a vector group): $\boldsymbol{\delta} = \mathbf{y}_k - h(\mathbf{X})$.

4 Unscented Kalman Filter on Lie Groups

4.1 Augmented State

$$\boldsymbol{\eta} = (\boldsymbol{\varepsilon}^\top, \mathbf{q}^\top, \mathbf{r}^\top)^\top \in \mathbb{R}^L, \quad L = 2p + q = 33, \qquad \mathbf{P}_\eta = \mathrm{blkdiag}(\mathbf{P}, \mathbf{Q}, \mathbf{R}).$$

Weights ($\alpha = 0.014$, $\beta = 2$, $\kappa = 0$, $\lambda = \alpha^2(L+\kappa) - L$):

$$W_m^{(0)} = \frac{\lambda}{L+\lambda}, \quad W_c^{(0)} = \frac{\lambda}{L+\lambda} + (1 - \alpha^2 + \beta), \quad W_m^{(i)} = W_c^{(i)} = \frac{1}{2(L+\lambda)}, \; i \geq 1.$$

4.2 Sigma Points (`SigmaPointsLie.m`)

$\mathbf{S} = \mathrm{chol}(\mathbf{P}_\eta)^\top$ (lower triangular):

$$\boldsymbol{\xi}^{(0)} = \mathbf{0}_L, \quad \boldsymbol{\xi}^{(i)} = +\sqrt{L+\lambda}\,\mathbf{S}_{:,i}, \quad \boldsymbol{\xi}^{(i+L)} = -\sqrt{L+\lambda}\,\mathbf{S}_{:,i}, \quad i = 1, \ldots, L.$$

4.3 Prediction (`prediction_UKF_Lie.m`)

$$\begin{aligned}
\mathcal{X}_{k-1}^{(i)} &= \mathbf{X}_{k-1}\,\mathrm{Exp}(\boldsymbol{e}^{(i)}) && \text{(lift)} \\
\mathcal{X}_{k|k-1}^{(i)} &= \mathcal{X}_{k-1}^{(i)}\,\mathrm{Exp}(\Omega^{(i)}\Delta t + \mathbf{q}^{(i)}) && \text{(propagate)} \\
\bar{\mathbf{X}}_k &= \text{Frechet mean}\{\mathcal{X}_{k|k-1}^{(i)}, W_m^{(i)}\} && \text{(mean)} \\
\boldsymbol{\varepsilon}^{(i)} &= \mathrm{Log}(\bar{\mathbf{X}}_k^{-1}\mathcal{X}_{k|k-1}^{(i)}) && \text{(algebra errors)} \\
\mathbf{P}_{k|k-1} &= \sum_i W_c^{(i)}\boldsymbol{\varepsilon}^{(i)}(\boldsymbol{\varepsilon}^{(i)})^\top + \mathbf{Q} && \text{(covariance)}
\end{aligned}$$

Frechet mean iteration (`media_nula_g.m`):

$$\bar{\mathbf{X}}^{(j+1)} = \bar{\mathbf{X}}^{(j)}\,\mathrm{Exp}\Big(\alpha^2 \sum_i W_m^{(i)} \mathrm{Log}\big((\bar{\mathbf{X}}^{(j)})^{-1}\mathcal{X}^{(i)}\big)\Big)$$

until $\|\boldsymbol{\sigma}^{(j)}\| < 10^{-3}$ (max 30 iterations).

4.4 Update (`Update_UKF_Lie.m`)

$$\mathcal{Y}^{(i)} = \begin{pmatrix} \mathbf{I} & \mathbf{p}^{(i)} + \mathbf{C}^{(i)}\boldsymbol{\ell}_b + \mathbf{r}^{(i)} \\ \mathbf{0} & 1 \end{pmatrix} \qquad \text{(meas sigma pts)}$$

$$\bar{\mathbf{y}} = \alpha^2 \sum_i W_m^{(i)} \mathbf{y}^{(i)} \qquad \text{(predicted mean)}$$

$$\mathbf{P}_{hh} = \sum_i W_c^{(i)} \boldsymbol{\delta}^{(i)} (\boldsymbol{\delta}^{(i)})^\top + \mathbf{R} \qquad \text{(innovation cov)}$$

$$\mathbf{P}_{gh} = \sum_i W_c^{(i)} \boldsymbol{\varepsilon}^{(i)} (\boldsymbol{\delta}^{(i)})^\top \qquad \text{(cross cov)}$$

$$\mathbf{K} = \mathbf{P}_{gh}\mathbf{P}_{hh}^{-1} \qquad \text{(Kalman gain)}$$

$$\mathbf{X}_{k|k} = \bar{\mathbf{X}}_k\,\mathrm{Exp}(\mathbf{K}(\mathbf{y}_k - \bar{\mathbf{y}})) \qquad \text{(state update)}$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}\,\mathbf{P}_{gh}^\top \qquad \text{(covariance update)}$$

5 Workflow and Pipeline

5.1 High-Level Architecture

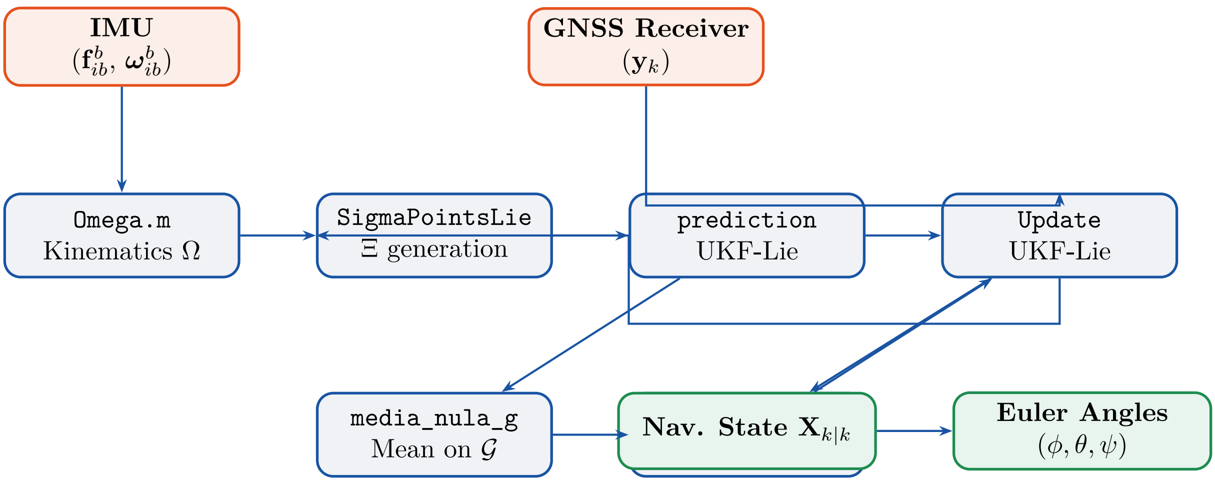

Figure 2: High-level architecture and data flow of the UKF-Lie system.

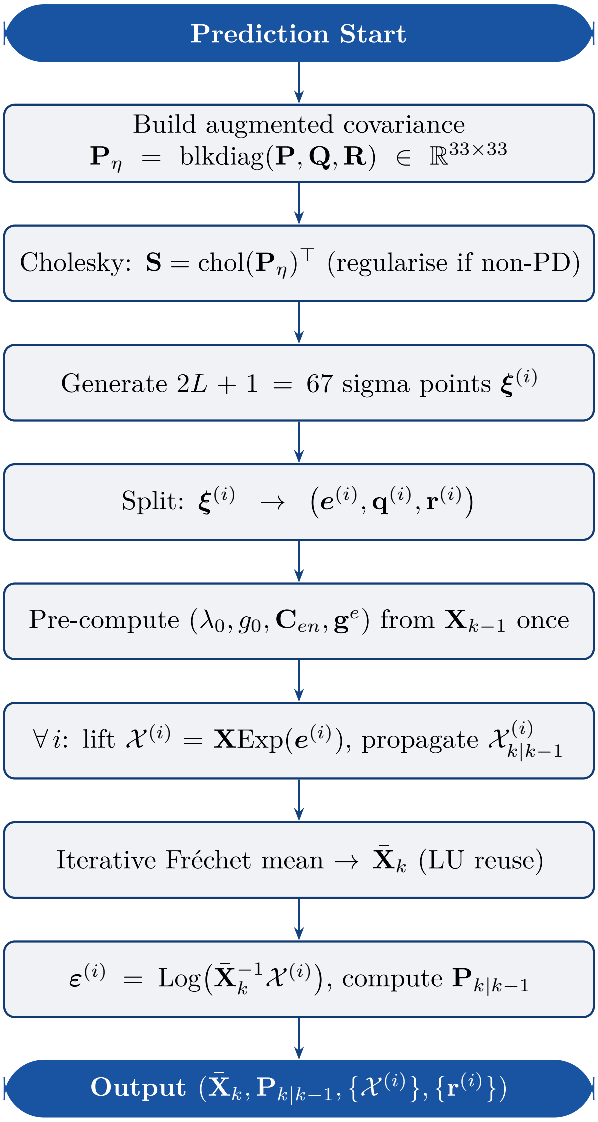

Figure 3: Prediction step flowchart.

5.2 Prediction Flowchart

5.3 Update Flowchart

5.4 Timing Diagram

5.5 Sigma-Point Geometry

6 Implementation Details

6.1 Key Functions

6.2 Initialisation

$$\mathbf{X}_0 = \mathrm{blkdiag}\left(\begin{pmatrix} \mathbf{C}_{eb}^0 & \mathbf{0} & \mathbf{p}_0 \\ \mathbf{0} & \mathbf{I} & \end{pmatrix}, \mathbf{I}_4, \mathbf{I}_4\right), \tag{1}$$

$$\mathbf{P}_0 = \mathrm{diag}(10^{-6}\mathbf{1}_3^\top,\ 10^{-9}\mathbf{1}_3^\top,\ 10^{-6}\mathbf{1}_6^\top,\ 10^{-9}\mathbf{1}_3^\top). \tag{2}$$

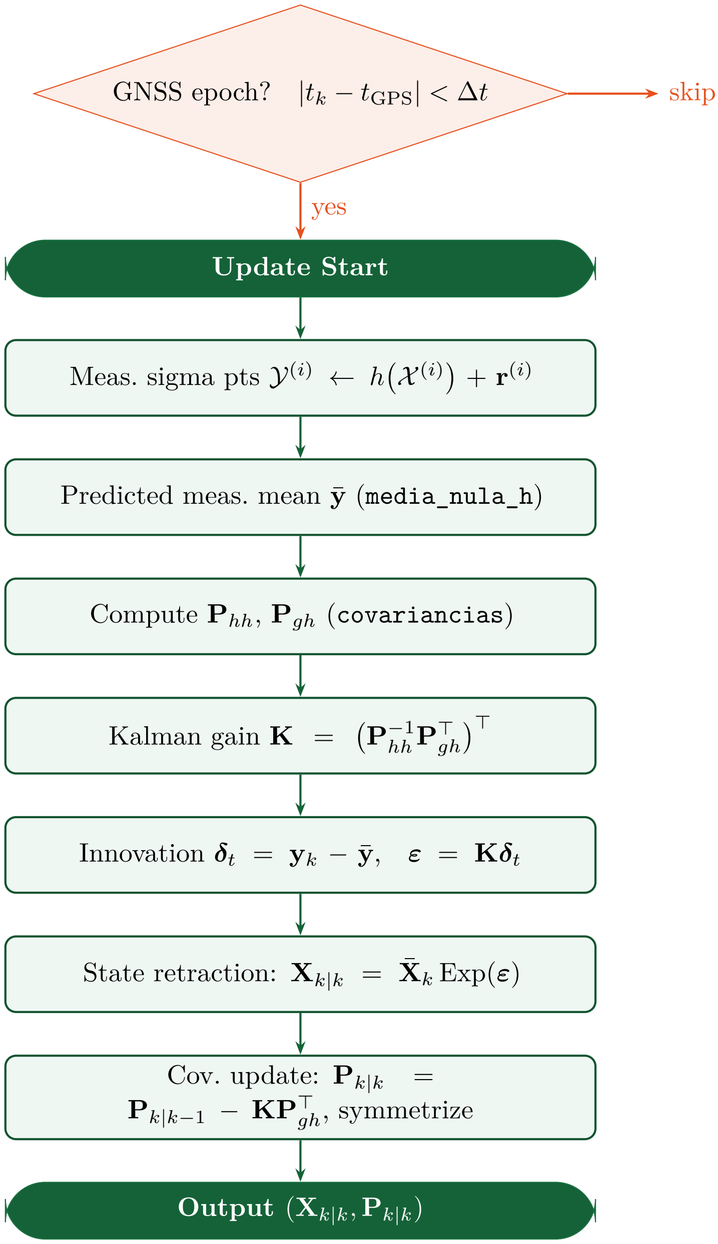

Figure 4: Update step flowchart.

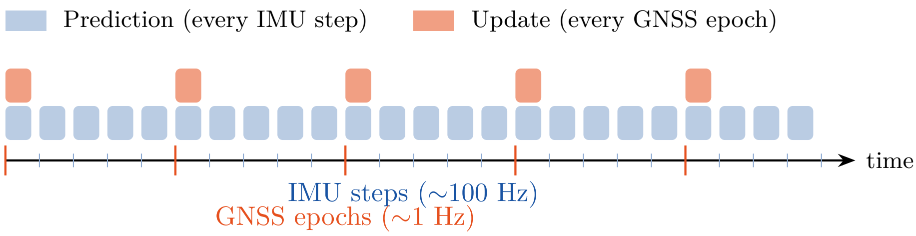

Figure 5: Timing of predict/update steps in `run_UKF_Lie.m`.

6.3 Noise Parameters

6.4 Optimisations

1. **Closed-form** Log: replaced `eig()` with Rodrigues extraction — 5–10× faster.

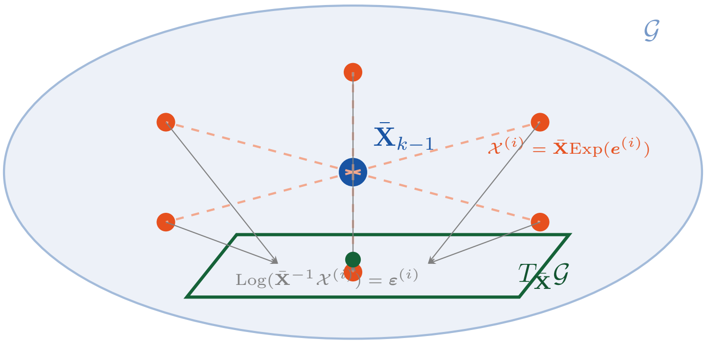

Figure 6: Sigma points on $\mathcal{G}$ and projection to tangent space $T_{\bar{\mathbf{X}}}\mathcal{G}$ via Log.

Table 2: Key MATLAB functions and their roles.

Function	Input/Output	Description
`SigmaPointsLie`	$\to (\Xi, W_m, W_c)$	67 sigma pts via Cholesky of $\mathbf{P}_\eta$
`prediction_UKF_Lie`	$\to (\bar{\mathbf{X}}, \mathbf{P}, \{\mathcal{X}^{(i)}\}, \{\mathbf{r}^{(i)}\})$	Full prediction
`Update_UKF_Lie`	$\to (\mathbf{X}_{k\|k}, \mathbf{P}_{k\|k})$	Full update
`media_nula_g`	$\to \bar{\mathbf{X}}$	Frechet mean on $\mathcal{G}$
`media_nula_h`	$\to \bar{\mathbf{Y}}$	Mean on $T(3)$
`covariancias`	$\to (\mathbf{P}_{hh}, \mathbf{P}_{gh})$	Innovation/cross-cov
`exp_multiSE23T6`	$\mathbb{R}^{15} \to \mathcal{G}$	Group exponential
`log_multiSE23T6`	$\mathcal{G} \to \mathbb{R}^{15}$	Group logarithm
`log_multiSE3`	$\mathrm{SE}_2(3) \to \mathbb{R}^9$	Rodrigues-based log
`Omega`	$(\mathbf{X}, \mathbf{u}) \to \mathbb{R}^{15}$	IMU kinematics
`SingleLlaFromEcef`	$\mathbb{R}^3 \to (\lambda, \ell, h)$	ECEF→geodetic
`gravityModel`	$\lambda \to g$	WGS-84 gravity

Table 3: Filter noise standard deviations.

Parameter	Symbol	Value	Units
Gyroscope	σ_g	10^{-3}	rad/s
Accelerometer	σ_a	10^{-3}	m/s^2
Velocity noise	σ_v	$10^{-1.5}$	m/s
Gyro bias	σ_{b_g}	$10^{-4.5}$	rad/s
Acc. bias	σ_{b_a}	$10^{-4.5}$	m/s^2
GNSS position	σ_{GPS}	10^{-1}	m

2. **Geodetic pre-computation**: `SingleLlaFromEcef` called once/step, not 67 times.
3. **LU reuse**: factorize g once per mean iteration, reuse for all sigma-point solves.
4. **Cholesky PSD check**: `chol()` replaces `eig()` for PD verification.
5. **Minor**: pre-transpose `Cen`, trace via diagonal indexing.

7 Evaluation

RMSE metrics:

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{N}\sum_k \|\hat{\mathbf{p}}_k - \mathbf{p}_k^{\mathrm{ref}}\|^2}, \quad \mathrm{RMSE}_\psi = \sqrt{\frac{1}{N}\sum_k (\hat{\psi}_k - \psi_k^{\mathrm{ref}})^2}.$$

Uncertainty monitor: $\tau_k = \mathrm{tr}\Big(\mathbf{P}_{k|k}\Big)$.
Datasets: `rectangular`, `circular`, `helicoidal`.

A Code Cross-Reference

Table 4: Equation-to-code mapping.

Equation	Section	MATLAB file
Exp on $\mathrm{SE}_2(3)$	2.2	`exp_multiSE3.m`
Exp on $\mathcal{G}$	2.3	`exp_multiSE23T6.m`
Log on $\mathrm{SE}_2(3)$	2.2	`log_multiSE3.m`
Log on $\mathcal{G}$	2.3	`log_multiSE23T6.m`
$\mathrm{Ad}_{\mathbf{X}}$	2.3	`Ad_G.m`
$[\xi]^\wedge$ bracket	2.2	`bracketUp_SE_23.m`
$\mathbf{C}_{en}$	3.1	`DCM_en.m`
$g(\lambda)$ WGS-84	3.1	`gravityModel.m`
ECEF→LLA	3.1	`SingleLlaFromEcef.m`
$\Omega(\mathbf{X}, \mathbf{u})$	3.2	`Omega.m`
Sigma points	4.2	`SigmaPointsLie.m`
Propagation	4.3	`prediction_UKF_Lie.m`
Frechet mean	4.3	`media_nula_g.m`
$\mathbf{P}_{k\|k-1}$	4.3	`prediction_UKF_Lie.m`
Meas. mean	4.4	`media_nula_h.m`
$\mathbf{P}_{hh}, \mathbf{P}_{gh}$	4.4	`covariancias.m`
$\mathbf{K}$, update	4.4	`Update_UKF_Lie.m`
Main loop	5	`run_UKF_Lie.m`